Vishal Jeyam

Fayetteville, AR, 72704 | jeyamvishal5@gmail.com | linkedin.com/in/vjeyam/ | vishaljeyam.com

Education

Master of Science in Computer Science | University of Arkansas Expected Graduation: December 2027

- **Thesis:** Improving a stock movement prediction model by incorporating news or sentiment data alongside traditional market data

Bachelor of Science in Computer Science | University of Arkansas August 2021 to May 2025

Work Experience

Research Assistant | Smart Agricultural & Food Engineering Lab May 2025 to Present

- Operated a Clearpath A200 Husky autonomous mobile robot, integrating ROS-based workflows for sensor fusion, mapping, and navigation as part of an AI-powered environmental monitoring and sanitation system
- Collected and curated large-scale multimodal datasets spanning LiDAR, vision, odometry, and position data across poultry facility environments to support computer vision modeling
- Trained a YOLOv11 segmentation model to classify yellow vs. green plants in real time, achieving *98.1%* precision and *92.6%* mAP@50 on validation, with *98.2%* accuracy on unseen test data, powering the navigation decisions that led to *2nd of 7* teams in both the autonomous navigation and identification categories

Undergraduate Research Intern | Washington State University June 2024 to August 2024

- Created a real-time crop health and yield prediction pipeline for a WSU wheat breeding program (REEU), using a provided dataset of *7,000+* paired NIR and RGB multispectral images
- Fine-tuned a YOLOv8 model to detect and segment wheat field plots, achieving *82%* detection accuracy and *84%* segmentation accuracy, to isolate crop regions for downstream analysis
- Computed vegetation indices (Soil Crop Index, Green Normalized Difference Vegetation Index) within segmented plots to predict wheat health and yield, reducing the cost and manual labor of traditional breeding-program assessment

Software Engineer Intern | Arkansas Blue Cross and Blue Shield June 2023 to August 2023

- Built a proof-of-concept Kafka consumer client in Python to receive real-time data from an existing producer pipeline, validating end-to-end delivery by confirming a test dataset of team member names published by the producer was correctly received by the consumer
- Developed a proof-of-concept Elasticsearch ingestion script to repeatedly pull data from one or more PostgreSQL databases and consolidate it into a single unified database, enabling cross-team access to previously siloed data
- Contributed to the Integration Hub team's pipeline lifecycle, spanning stream ingestion with Kafka to search indexing with Elasticsearch, across distributed systems

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Projects

LLM Model Optimization August 2026

- Benchmarked Qwen2.5-7B-Instruct across 4 quantization level (F16, Q8_0, Q4_K_M, Q2_K) in Python and Ollama, measuring accuracy, latency, and throughput on MedQA and PubMedQA datasets
- Reduced model size *69%* (14.19 GB to 4.36 GB) at Q4_K_M while improving MedQA accuracy from *58.8%* to *59.1%* and PubMedQA accuracy from *74.0%* to *75.1%*
- Increased MedQA inference throughput *87%* (82.5 to 154.0 tokens/sec) at Q4_K_M with no accuracy loss, while heavier Q2_K compression dropped MedQA accuracy *9 points*

Skills

- Programming Languages: Python, SQL, C++, Bash, JavaScript/Typescript, HTML/CSS
- ML & AI: PyTorch, Scikit-learn, Hugging Face, Ollama, Deep Learning, Machine Learning, LLMs
- Computer Vision: OpenCV, Ultralytics/YOLO, Torchvision, Object Detection, Image Segmentation
- Data & Databases: Pandas, NumPy, Kafka, Elasticsearch, PostgreSQL, MySQL, Supabase
- Cloud & Tools: Docker, Azure, Git/Github, Jupyter, Matplotlib, Seaborn, Tableau